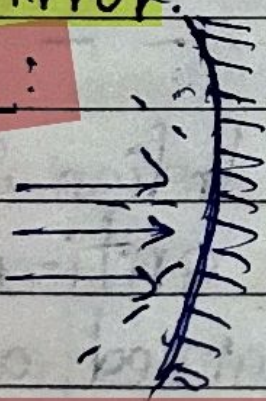


Ch-10

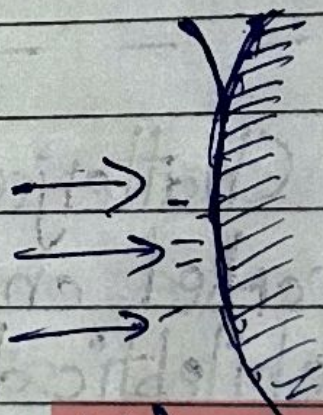
Spherical mirrors are a common type of curved mirror, which are shaped like a part of hollow glass sphere. So, simply, mirrors whose reflecting surfaces are spherical, are called spherical mirrors.

The spherical mirror whose surface is curved inwards is called a concave mirror. The spherical mirror whose surface is curved outward is called a convex mirror.

Fig:



(a) Concave



(b) Convex

Spherical mirrors are not made by cutting them from a hollow glass sphere but by grinding & polishing a flat glass into a curved surface. The reflective coating (Eg: Aluminium's thin layer) is applied on curved surface to make the convex or concave mirror.

Mirrors

DATE: _____
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Concave

When object is placed close to the mirror, it appears enlarged & erect.

As you keep moving the obj. farther away from mirror, the image of obj keeps getting smaller.

When object is kept farther from mirror, it appears inverted.

Uses: Reflector of torch, by a dentist

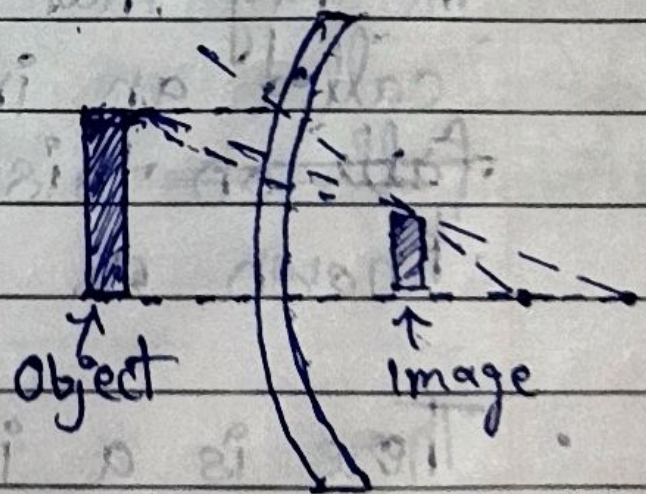
Convex

The image of the obj. appears diminished & erect (always)

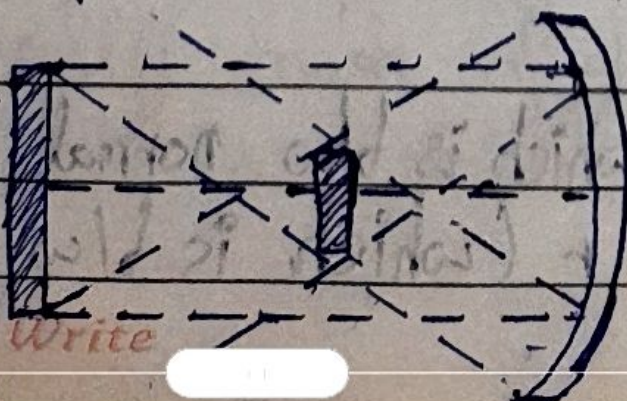
The size of image also becomes smaller as obj. gets farther.

Uses: Side view mirror, Surveillance mirror

Dia. (Not in syllabus):

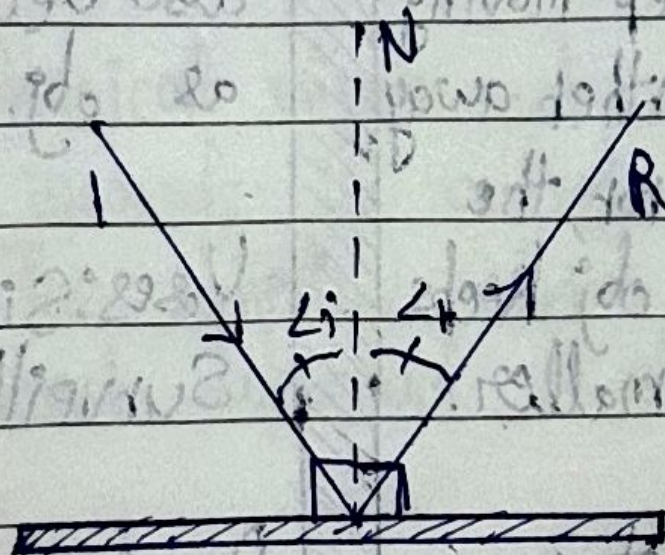


Dia. for concave (Not in syllabus):



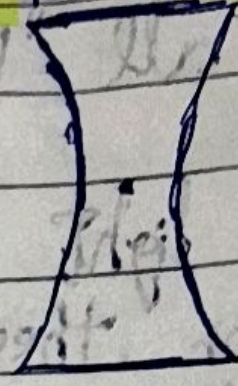
A plane mirror is a type of mirror which forms an image of the same ~~size~~ size of the object. The image is erect but laterally inverted.

Lateral inversion can be seen in all 3 types of mirror.



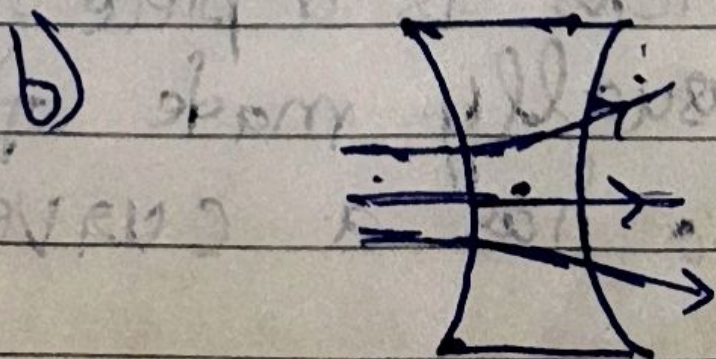
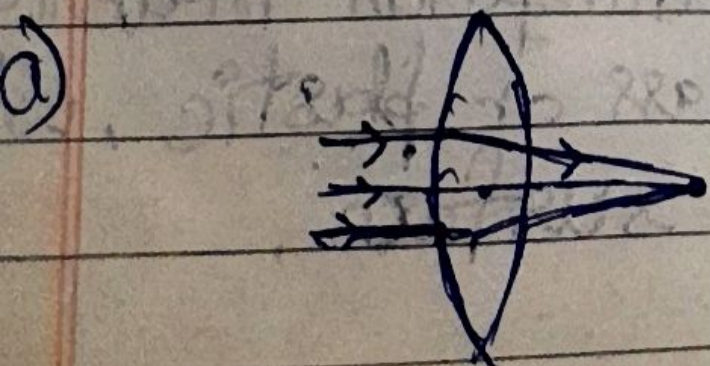
- The ray that falls on the mirror is called an incident ray. The ray that ~~falls on~~ is reflected by the mirror is known as reflected ray.
- There is an imaginary line drawn perpendicular to the point where the incident ray strikes the mirror. It is called normal.
- The angle i (which is b/w normal & incident ray) & angle r (which is b/w normal & reflected ray).

- The lens which is thicker at middle as compared to its edges, is a convex lens. But, the lens which has a thicker edges than the middle, is a concave lens. Diagram



- When an obj. is placed behind a convex lens at a small distance, the obj. appears enlarged & erect in size. But, as the obj. move farther, it starts to diminish & become inverted. When the obj. is placed behind a concave lens, it always appears diminished & erect. The size of image changes, as the dis. b/w obj. & lens increases.

- Convex lens converges light rays & concave lens diverges light rays.



- Uses - Convex : Magnifying glass, eyes of humans, telescope

- Concave : Door peepholes, Flashlights